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Cott-ADNet: Lightweight Real-Time Cotton Boll and Flower Detection Under Field Conditions

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Why Cotton Detection Matters

Cotton supports 100+ million farmers worldwide.
However harvesting still relies on:

- Labor-intensive manual picking.
- Low efficiency and high cost.
- Missed optimal harvest window → yield loss.

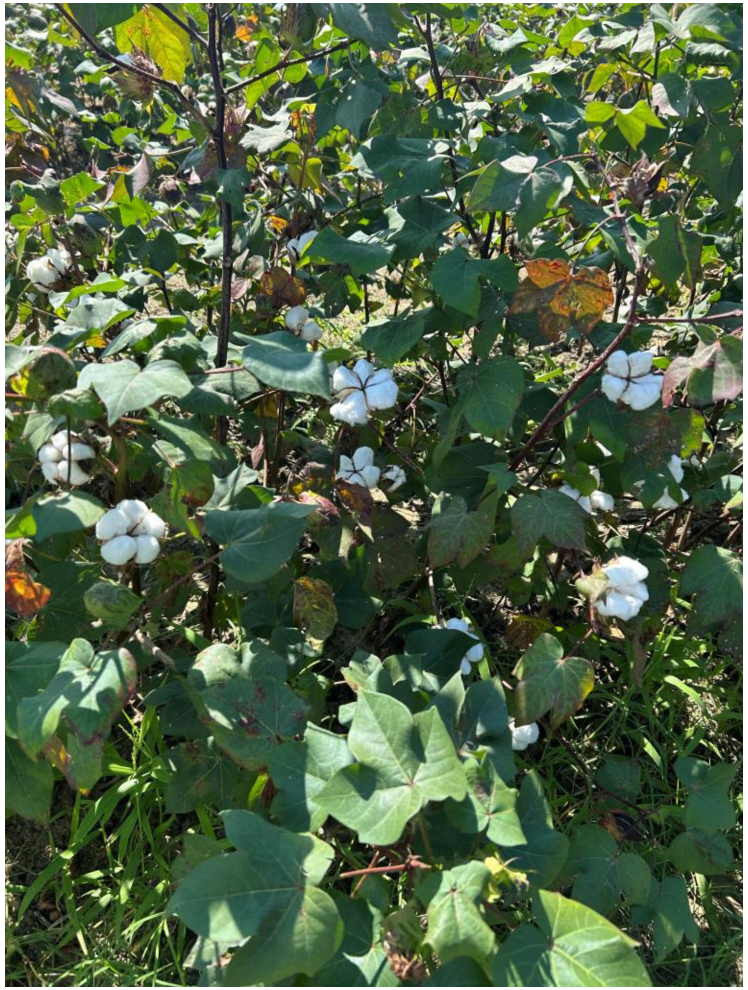
Automated cotton perception is critical for:

- Robotic harvesting.
- Yield estimation.
- Smart breeding.

Challenges in Real Fields

Real-world cotton detection is difficult due to:

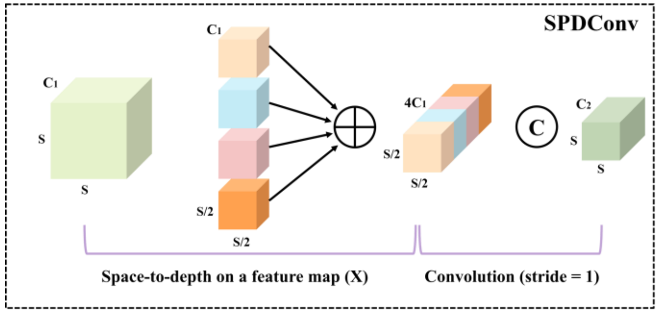
- Dense and occluded bolls.
- Small object size.
- Illumination variation.
- Background clutter.
- Lack of real field validation.

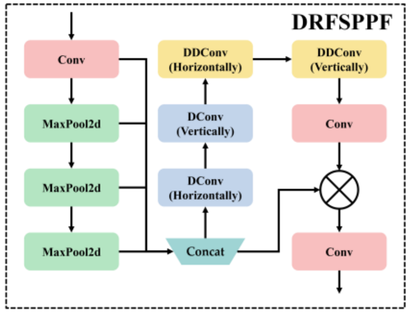


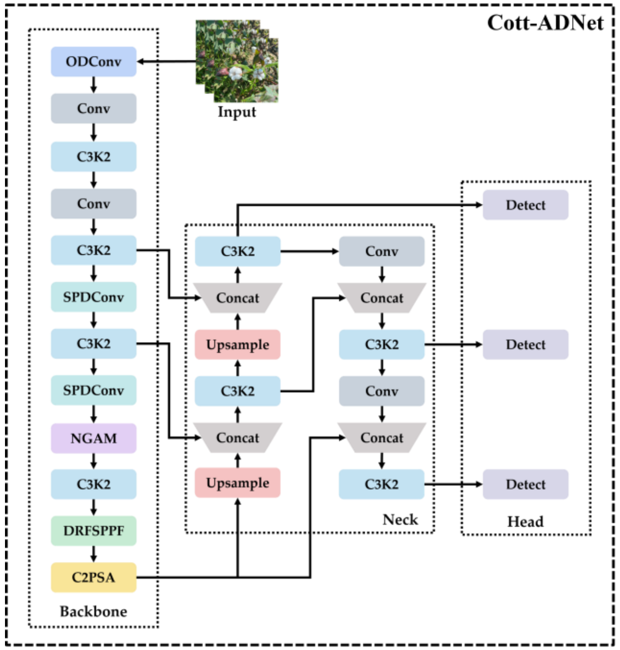
We need a detector that is:

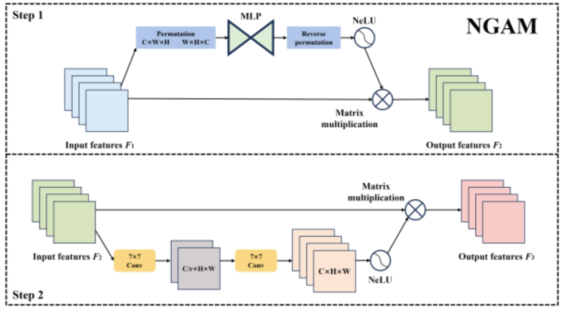
- Accurate + Lightweight + Deployable.

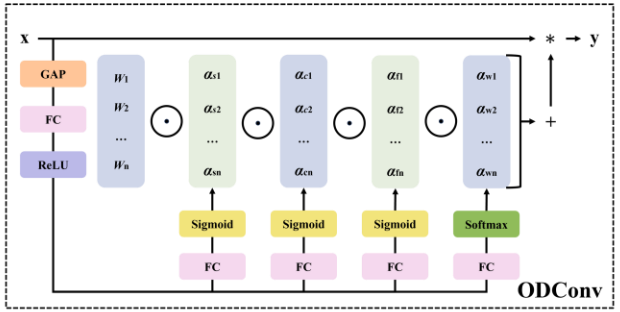
Our Solution











Cott-ADNet

- A lightweight detector built on YOLOv11n.

Key Contributions

- Task-driven lightweight architecture refinement.
- NGAM for weak feature modeling.
- DRFSPPF for large receptive fields.
- Real farm external validation dataset.

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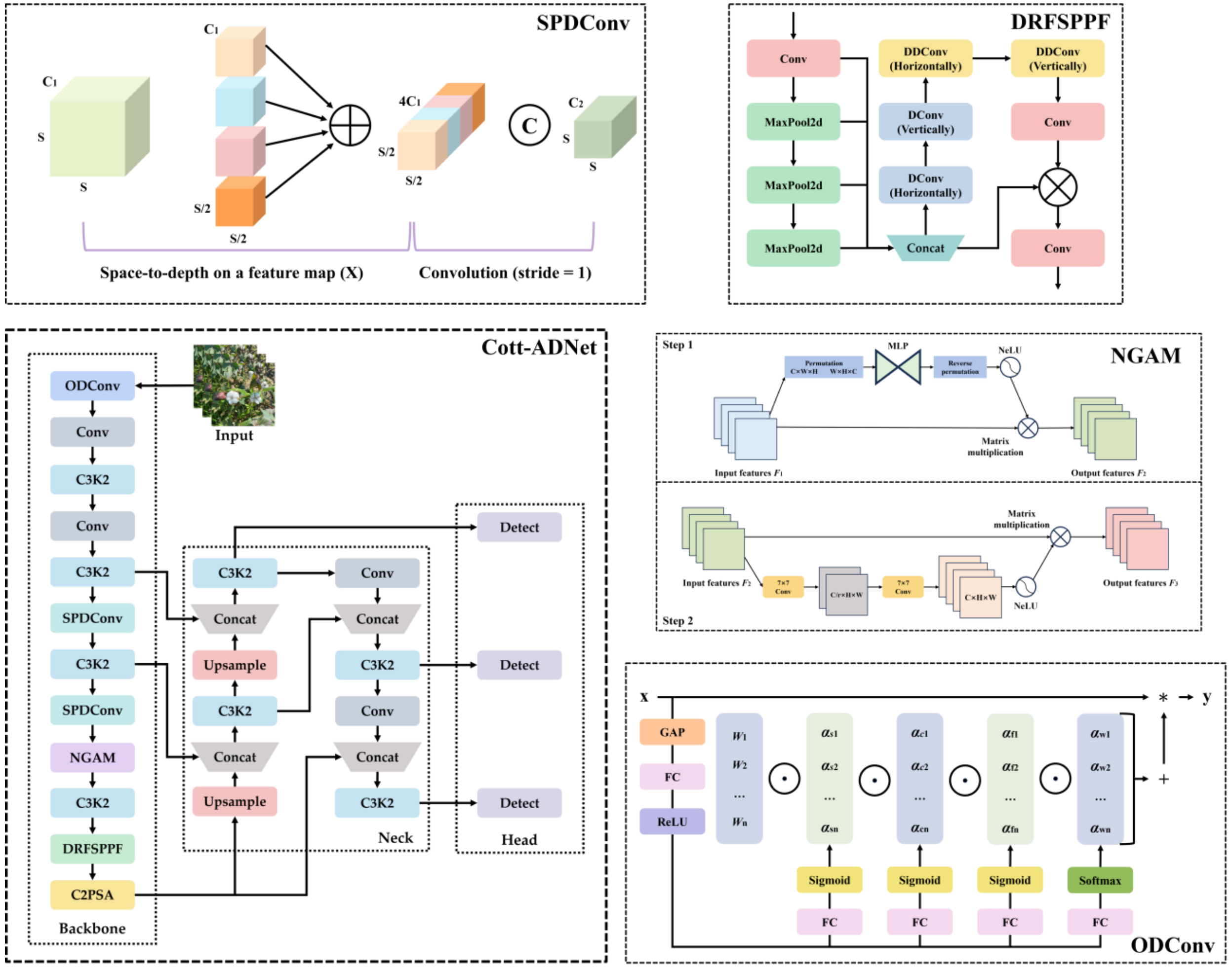
Overall Architecture



Cott-ADNet Architecture

Key improvements:

- ODConv → dynamic convolution.
- SPDConv → lossless downsampling.
- NGAM → global attention enhancement.
- DRFSPPF → enlarged receptive field.

Our **final** model achieves strong accuracy with only 7.5 GFLOPs.

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NGAM: NeLU-Enhanced Global Attention

Problem:

- ReLU/Sigmoid → gradient saturation & dying neurons.

Solution:

- Replace activations with NeLU

Benefits:

- Preserves gradient for negative inputs
- Enhances weak-texture feature modeling
- Improves training stability and generalization

$$\mathrm{NeLU}(x) = \begin{cases} x, & x > 0 \\ -\dfrac{\alpha}{1 + x^2}, & x \leq 0 \end{cases}$$

$$\frac{d}{dx}\mathrm{NeLU}(x) = \begin{cases} 1, & x > 0 \\ \frac{2\alpha x}{(1+x^2)^2}, & x \leq 0 \end{cases}$$

DRFSPPF: Enlarged Receptive Field Modeling

Goal:

- Capture small objects + large context simultaneously.

Key design:

- Multi-scale pooling.
- Large-kernel depthwise conv.
- Dilated depthwise conv.
- Residual recalibration.

Result:

- Better multi-scale context with a small extra computational cost.

Algorithm 1 Dilated Receptive Field SPPF (DRFSPPF)

Require: Feature map $F^C \in \mathbb{R}^{H \times W \times C}$

Ensure: Enhanced feature representation DRFSPPF(F^C)

1: **Multi-scale pooling:** Apply pooling with kernels $\{5, 9, 13\}$ and concatenate:
$$F^{C'} = \text{Concat}(F^C, P_5(F^C), P_9(F^C), P_{13}(F^C))$$

2: **Large-kernel DConv:** Apply depthwise conv ($k = 11$):
$$F^C_{\text{DConv}} = W^C_{(2d-1)\times 1} * (W^C_{1\times(2d-1)} * F^{C'})$$

3: **Dilated DConv:** Expand receptive field with dilation d:
$$F_{\text{DDConv}} = W^C_{[11/d]\times 1} * (W^C_{1\times[11/d]} * F^C_{\text{DConv}})$$

4: **Residual recalibration:** Fuse with F^C:
$$\bar{F}^C = (W_{1\times 1} * F_{\text{DDConv}}) \odot F^C$$

5: **Output projection:** Final representation:
$$\text{DRFSPPF}(F^C) = W^{out}_{1\times 1} * \bar{F}^C$$

Datasets

Training Dataset:

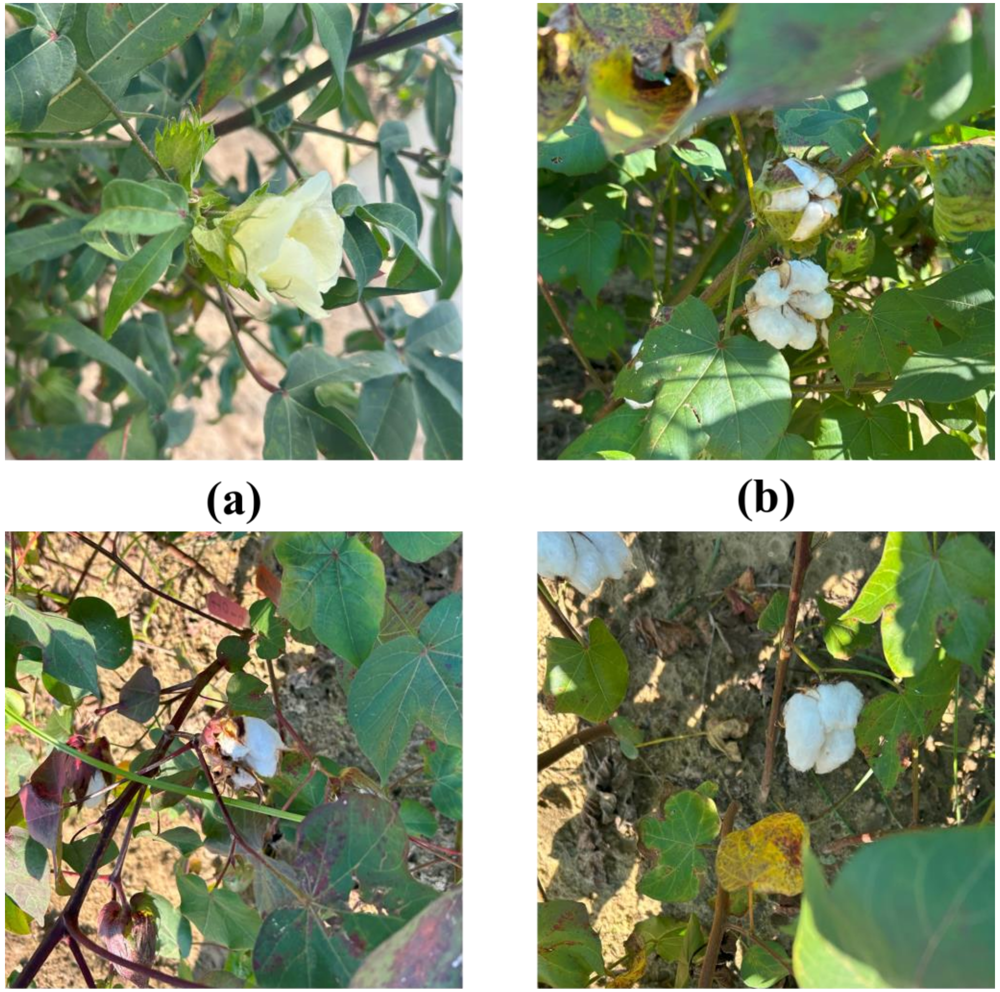
- 4,966 images
- Multi-source collection
- Expert verified labels

External Field Dataset:

- 1,216 real farm images
- Used only for evaluation

Target Classes:

- Flower • Partly opened • Defected • Fully opened



(a) (b) (c) (d)

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Comparison Experiments

Comparison with SOTA Detectors

Cott-ADNet achieves:

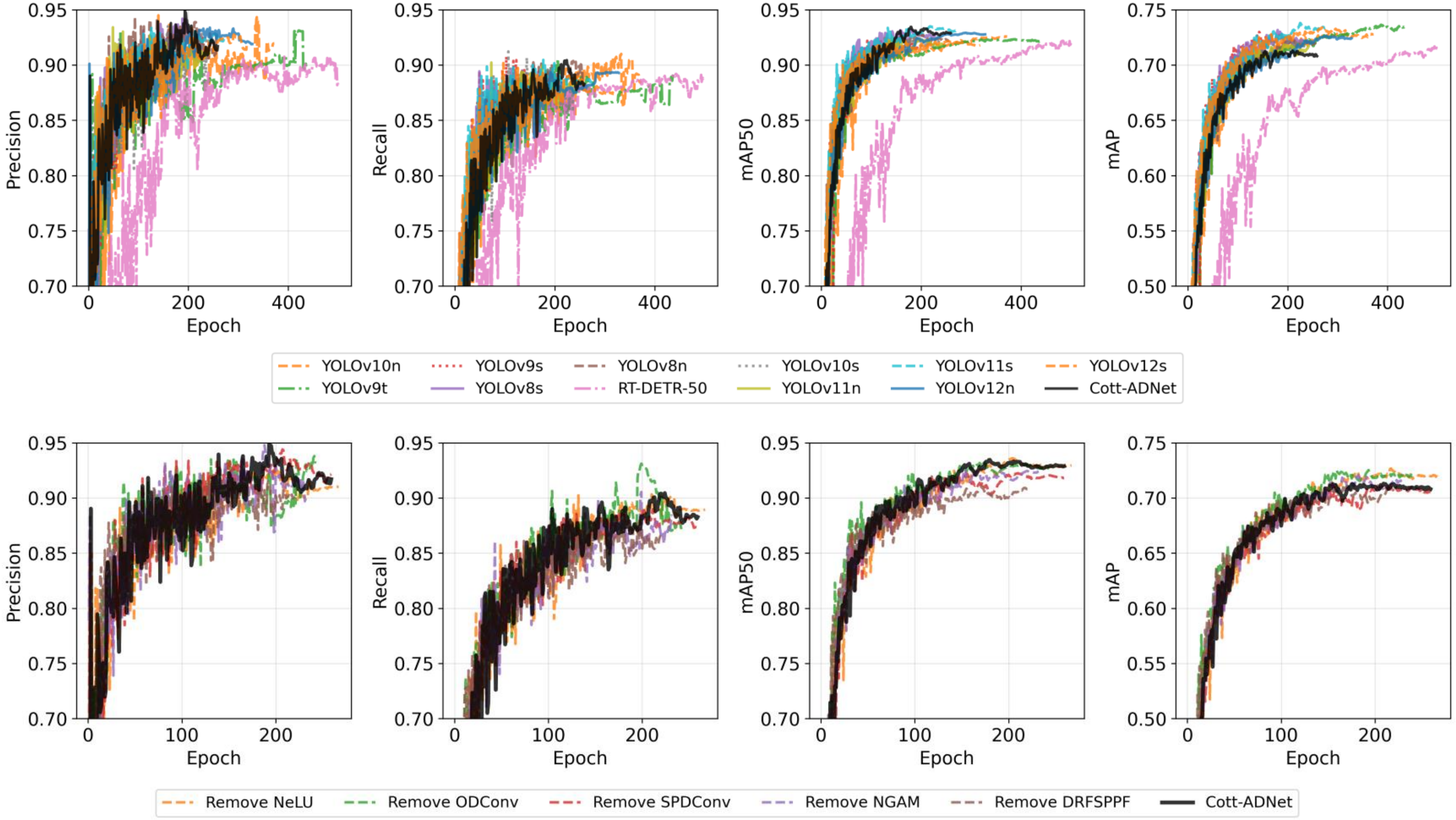
- Highest F1: 90.6%
- mAP50: 93.3%
- Only 7.5 GFLOPs

Compared to YOLOv11s:

- Similar accuracy with 3× lower computation

Model	P (%)	R (%)	F_1 (%)	mAP50 (%)	mAP (%)	GFLOPs
YOLOv8s	90.271	**89.516**	89.892	92.053	72.214	28.8
YOLOv9s	89.997	89.176	89.585	91.917	72.983	27.6
YOLOv10s	92.230	87.371	89.790	92.581	72.451	24.8
YOLOv11s	**92.615**	88.204	**90.395**	**93.417**	**73.658**	21.6
YOLOv12s	92.320	87.596	89.945	92.014	73.374	21.5
RT-DETR-50	91.166	88.333	89.727	91.672	71.805	130.5
YOLOv8n	**93.577**	86.693	90.022	92.308	71.674	8.9
YOLOv9t	90.313	88.109	89.197	92.317	**73.637**	8.5
YOLOv10n	90.026	**90.579**	90.289	92.416	72.825	8.4
YOLOv11n	90.236	88.224	89.227	92.197	71.844	6.4
YOLOv12n	92.654	88.285	**90.466**	**92.795**	72.640	**6.5**
Remove ODConv	92.334	87.097	89.639	92.571	72.527	7.6
Remove SPDConv	**93.433**	87.970	90.619	92.199	71.202	7.9
Remove DRFSPPF	90.486	86.211	88.297	90.925	71.068	7.3
Remove NGAM	90.232	87.399	88.793	92.045	72.068	**6.2**
Remove NeLU	91.736	88.885	90.288	93.077	**72.664**	7.5
Cott-ADNet	91.543	**89.753**	**90.639**	**93.285**	71.296	7.5

Ablation Study



Key findings:

- Removing NGAM → accuracy drops.
- Removing DRFSPPF → large mAP decrease.
- All modules contribute to final performance.

Real Field Detection

Strong generalization in real farm scenes.

Failure cases:

(1) Severe occlusion. (2) Strong shadows. (3) PB vs FB confusion.

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Conclusion

Cott-ADNet is:

- Lightweight (7.5 GFLOPs)
- Accurate and robust
- Ready for real deployment

Applications:

- Robotic harvesting • Yield monitoring • Phenotyping

Future Work:

- UAV/UGV deployment • Multi-modal sensing

Team & Collaborators

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Thank You!